

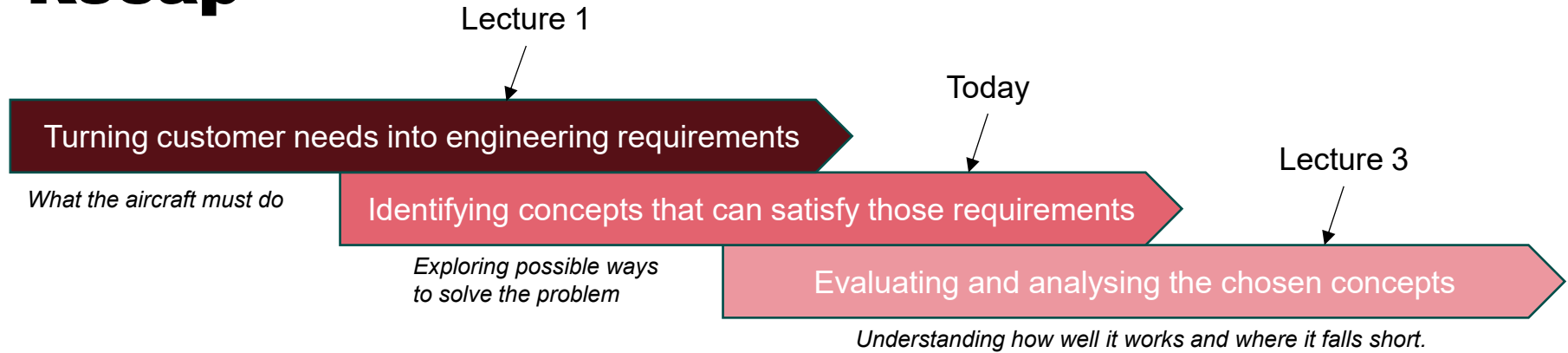
Aircraft Design 101

How do we choose a concept under uncertainty?

CADEM0016

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Recap

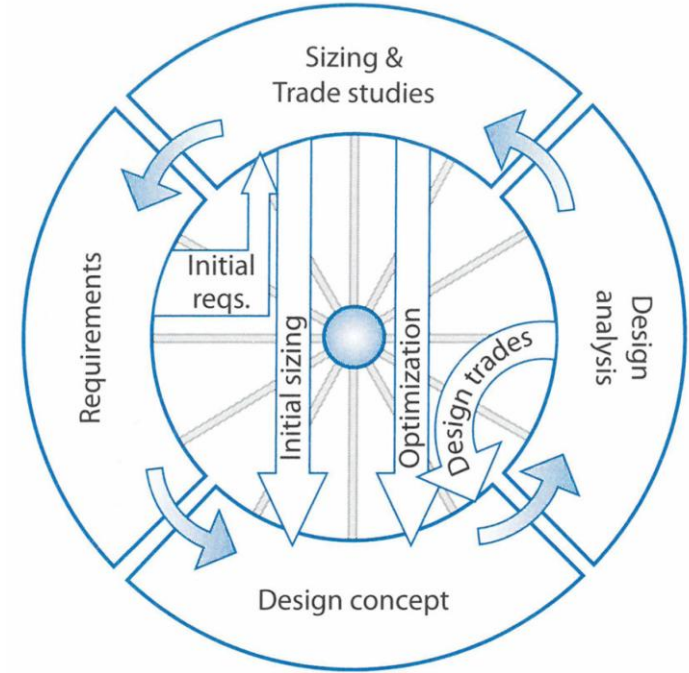


- Customer requirements → TLARs
- TLARs define the problem
- Today: how to choose concepts that **are likely** to solve that problem

The Aircraft Design Process

The Design Process

- Aircraft design is an iterative process
 - But iteration is expensive and time-limited
- There is no single recipe for the “right” answer
 - Early choices strongly constrain later options
- Not all concepts deserve equal analysis effort
 - Poor ideas should be eliminated early
- TLARs define the feasible design space
 - e.g. short runways eliminate heavy, low-lift concepts



Lots of concepts in

-
- TLARs
- Conceptual
- Preliminary
- Detailed
- Design concept
- Design analysis
- Solving & Trade studies
- Sizing

The diagram shows a progression of three design stages. At the top is a 3D wedge shape labeled 'Conceptual design'. A blue arrow points down from it to a 3D rectangular block labeled 'Preliminary design'. Another blue arrow points down from the rectangular block to a detailed 3D rectangular block labeled 'Detail design'. This detailed block shows internal features like circles and squares, representing a more refined and specific design.

Design Phases

▪ Conceptual design

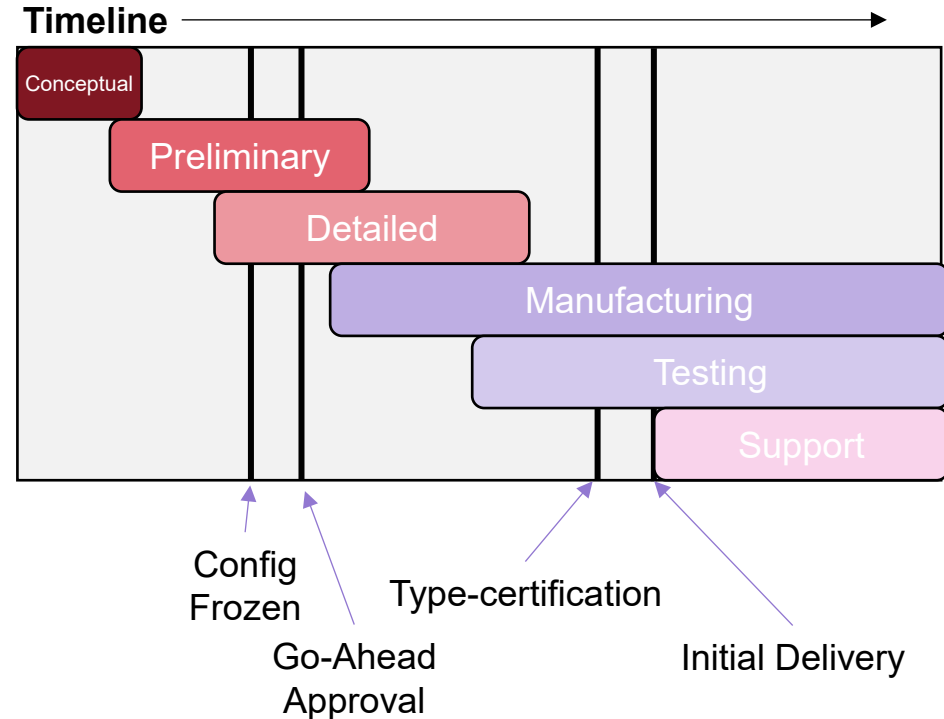
- Explore many configurations quickly.
- Define the overall architecture, evaluate trade-offs, and down-select to a single promising concept.

▪ Preliminary Design

- Develop the chosen concept in depth.
- Replace assumptions with higher-fidelity analysis
- size major components
- resolve integration issues
- confirm the design is technically and economically viable."

▪ Detailed Design

- Fully define every part of the aircraft.
- Produce manufacturing drawings
- certify structures and systems
- prepare the design for production and testing.



Design Phases

- **Conceptual design**

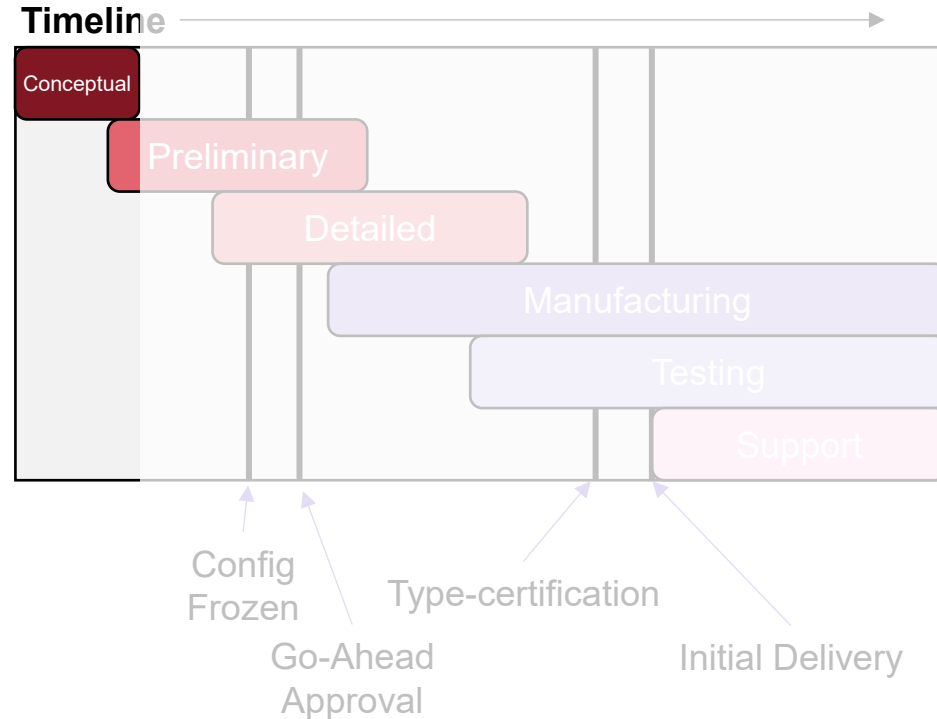
- Explore many configurations quickly.
- Define the overall architecture, evaluate trade-offs, and down-select to a single promising concept.

- **Preliminary Design**

- Develop the chosen concept in depth.
- Replace assumptions with higher-fidelity analysis
- size major components
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- confirm the design is technically and economically viable."

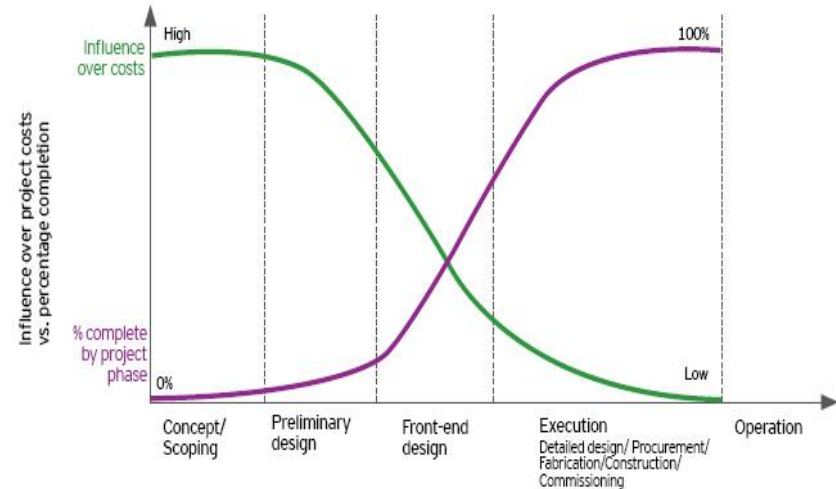
- **Detailed Design**

- Fully define every part of the aircraft.
- Produce manufacturing drawings
- certify structures and systems
- prepare the design for production and testing.



Design Phases

- **Conceptual design**
 - Explore many configurations quickly.
 - Define the overall architecture, evaluate trade-offs, and down-select to a single promising concept.
- Typically, 80% of programme costs are determined by the decisions made in the conceptual stage of design
 - Planform, engines, material, unit cost etc...
- These decisions often need to be made with inadequate information:
 - Will a given technology be mature in time?
 - How much benefit will a new material actually deliver?
 - ... What is the weight benefit of CFRP vs. Metallic?
 - ... What is the cost of CFRP vs. Metallic?
 - Is a suitable engine available?



Terminology

- There is lots of strange jargon in the aerospace industry.
 - Like any foreign language you need to understand it to communicate. If you're not sure, ask.
- **Concept**
 - The fundamental architectural idea of the aircraft
 - Examples: Flying-wing, Canard, 3-surface a/c, "Conventional"
- **Configuration**
 - A specific physical arrangement of major components within a concept. Defines geometry, placement, and interfaces of primary elements
 - Examples: High-wing vs low-wing, under-wing vs rear-mounted engines, T-tail vs conventional tail
- **Design**
 - The overall engineering activity of creating an aircraft. Includes **concept selection**, **configuration definition**, **sizing**, trade studies, and iteration
- **Sizing**
 - The process of creating **a valid aircraft design** capable of meeting the set requirements. It may not be the best aircraft.
 - It can be performed with any tools and may involve engineering judgement and experience
- **Optimisation**
 - The process of creating **the best aircraft** to meet the set requirements based on one or more target functions (e.g. fuelburn, minimum operating cost).
 - Usually involves exploration of the wider design space, e.g. technical trade studies or numerical optimisation

Exploring Concepts & Configurations

The Task at Hand

- You have a very limited time to choose concepts to progress later in the unit
 - Teamwork, Brainstorming and concept down-selection are key
- You've probably never done anything like this before
 - ...It's going to be a steep learning curve, with lots of uncertainty. You'll make mistakes – that's OK.
- You've probably never worked in teams this big before
 - ...This will also be a steep learning curve!

Some* Different Concepts & Configurations



Canard



Twin Fuselage



Twin Boom



3-Surface Aircraft



Joined Wing



Biplane
(an un-joined wing?)



Spanloader



Flying Wing



Lifting Body



“Conventional”

Single fuselage
Low, aft-swept wing
Underwing engines



Delta Wing



Variable Sweep



Mid Wing



Forward Sweep



High Wing



Inverse Delta Wing



Rear Paired Engines (Side & High)



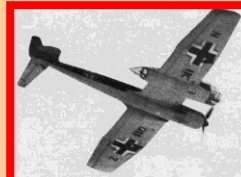
Rear Centreline Engines
(Buried in fuselage & fin-mounted)



Wing-mounted engines
(Over-wing, Mid-wing,
Buried-in-root & Tip-mounted)



Oblique Wing

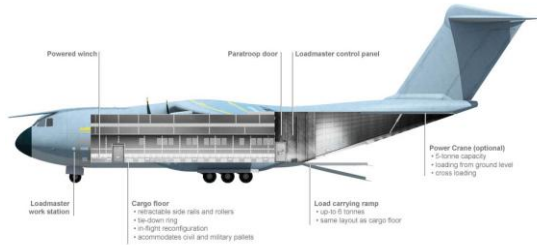


Asymmetric
everything but the
wing!



Thunderbird 2
(Podded payload,
Forward swept wing,
Nuclear engines)

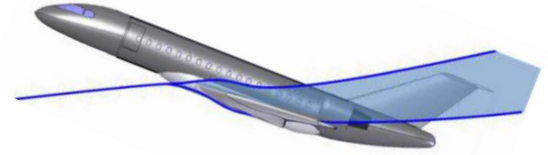
Cargo Examples



But what is the best Concept?

- It all depends what the aircraft is being asked to do!
 - Each concept/configuration has its good points
 - Each concept/configuration will also have its problems/issues
 - Some issues may be minor, Some issues may be show-stoppers
- The requirements should always drive the concept selection
 - Concepts that are “interesting” or “look nice” may be OK in the pre-selection stage, but your final choice should be based on the requirements
 - All previous aircraft programmes chose their configuration based on **their** set of requirements - Just because a configuration was suitable for them doesn't mean it will be suitable for you
 - Try to understand why they made the decisions they did
- You **will** be asked why you selected your concept/configuration
 - ... Make sure you have a good answer!

might be
Looks aren't important



Comparing Configurations

The Comparison Problem

- **Why intuition isn't enough**

- Early in design, everything looks possible. But different concepts excel in different areas, and human judgment alone can't reliably balance performance, cost, risk, and customer priorities.

- **Why we need structured evaluation**

- We need a transparent, repeatable way to compare concepts, expose trade-offs, and justify decisions.
- A structured method:
 - prevents bias
 - anchors decisions to the customers requirements
 - keeps the team aligned.

- To do this, we translate customer needs into weighted engineering criteria and score each concept consistently.

The Great Filter

- Analysis isn't cheap, first we must discard '*poor*' configurations
- Before scoring, we must first conduct feasibility screening
 - Can each concept meet the TLARs + any other hard constraints
- For example:
 - minimum cruise Mach number to meet mission criteria - can a concept achieve this?
 - Will technology be ready for Entry into service?

The Process

1. *“The Great Filter”*
2. Identify Customer Requirements (CRs)
3. Weight CRs
4. Identify Engineering Requirements (ERs)
5. Translate CR weightings to ERs
6. Assess concepts against weighted ERs

Identify Customer Requirements

- After reading the spec, you should be able to distil the needs of the customer into ~5-10 requirements, e.g.
 - Low operating cost
 - Low environmental impact
 - High reliability
 - Low turnaround times
 - Etc..
- These represent the “*voice of the customer.*”
- But the relative importance of these requirements is not all equal...

Ranking Customer Requirements

- When all requirements are weighted equally, minor “nice-to-have” features end up carrying the same influence as critical performance drivers like fuel burn, which distorts the decision
- There are multiple ways to rank / weight requirements
 - The analytic hierarchy process (AHP) is a robust example
 - Industry may use customer surveys, but for this project
- In your team, you can use whichever process you like, I will explain one called the Direct Point Allocation (PA).

Direct Point Allocation (PA).

- Each team member
 - is given 100 points
 - distributes their points across requirements relative to importance
- The mean score for each requirement is calculated across the team
- These scores are % importance, which can be changed into weights that sum to 1.

requirements	P1	P2	P3	P4	Sum	Mean	Weight
operating cost	40	50	35	33	158	39.5	0.40
environmental impact	40	40	55	34	169	42.25	0.42
reliability	20	10	10	33	73	18.25	0.18

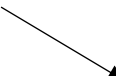
Identify Engineering Requirements

- Engineering requirements are the technical characteristics you will use to evaluate and compare design concepts
- What Makes a Good Requirement?
 - Measurable, Objective, Unambiguous, Relevant
- Examples:
 - MTOM, drag, maximum thrust, aircraft dimensions, fuel burn, noise level, wing area
- Note – try to avoid overlapping or redundant criteria (e.g., fuel efficiency vs. drag)

Translating CR weightings to ERs

- To convert weighted customer requirements to weighted Engineering requirements, we will use a process known as ***“House of Quality”***

Start by listing CRs + weights vertically



Customer Requirement	Weighting
operating cost	0.4
enviromental impact	0.42
reliability	0.18

Translating CR weightings to ERs

- To convert weighted customer requirements to weighted Engineering requirements, we will use a process known as ***“House of Quality”***

Listing ERs horizontally

Quality

ing ERs horizontally

							Favourability
Customer Requirement	Weighting	↘	↘	↗	↘	↘	
operating cost	0.4						
enviromental impact	0.42						
reliability	0.18						

Add a favourability note (e.g., is it good if this number is higher or lower)

Translating CR weightings to ERs

- To convert weighted customer requirements to weighted Engineering requirements, we will use a process known as ***“House of Quality”***

- 9 - Mass has **great** influence on *operating cost*
 3 - Mass has **weak** influence on *operating cost*
 0 - Mass has **no** influence on *operating cost*

		Mass	Drag	Lift Capability	Take-off Distance	TRL
Customer Requirement	Weighting	↘	↘	↗	↘	↘
operating cost	0.4	2	9	9	0	5
enviromental impact	0.42	3	8	1	0	0
reliability	0.18	1	0	0	2	8

Complete – interrelationship matrix
 score between 0 and 9

Translating CR weightings to ERs

- To convert weighted customer requirements to weighted Engineering requirements, we will use a process known as *“House of Quality”*

Calculate the importance of each engineering criteria on achieving customer requirements

		Mass	Drag	Lift Capability	Take-off Distance	TRL	
Customer Requirement	Weighting	↘	↘	↗	↘	↘	Favourability
operating cost	0.4	2	9	9	0	5	
enviromental impact	0.42	3	8	1	0	0	
reliability	0.18	1	0	0	2	8	
Score		2.24	6.96	4.02	0.36	3.44	17.02

$$\text{Purple Box} \times \text{Orange Box} = 0.4 \times 9 + 0.42 \times 8 + 0.18 \times 0 = 6.96$$

Translating CR weightings to ERs

- To convert weighted customer requirements to weighted Engineering requirements, we will use a process known as ***“House of Quality”***

- Normalise Scores to achieve a weighting for each engineering criteria

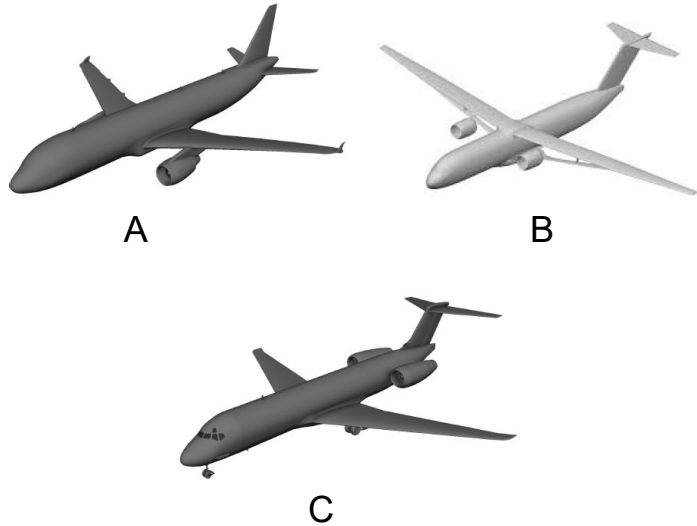
Quality

res to achieve a weighting for each
riteria

		Mass	Drag	Lift Capability	Take-off Distance	TRL		
Customer Requirement	Weighting	↘	↘	↗	↘	↘	Favourability	
operating cost	0.4	2	9	9	0	5		
enviromental impact	0.42	3	8	1	0	0		
reliability	0.18	1	0	0	2	8		
							Sum	
		Score	2.24	6.96	4.02	0.36	3.44	17.02
		Weighting	0.13	0.41	0.24	0.02	0.20	1

Multi-Criteria Decision Analysis (MCDA)

- Now that we have our weighting for each engineering criteria, we can compare different concepts

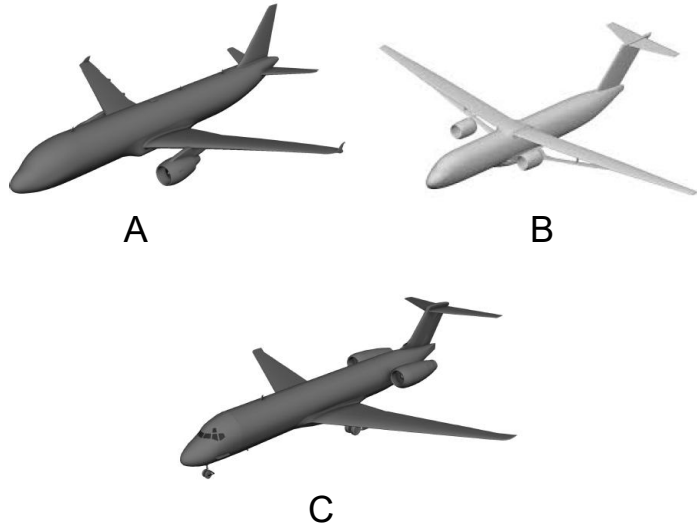


Criteria	Weighting	Concept		
		A	B	C
Mass	0.13	0	-2	0
Drag	0.41	0	3	1
Lift Capability	0.24	0	0	0
Take-off Distance	0.02	0	-1	0
TRL	0.20	0	-3	0
Total		0.00	0.34	0.41
Rank		3	2	1

Scoring	
5	Significantly better than TAW
2	Slightly better than TAW
0	Same as TAW
-2	Slightly worse than TAW
-5	Significantly worse than TAW

Multi-Criteria Decision Analysis (MCDA)

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		A	B	C
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Scoring	
5	Significantly better than TAW
2	Slightly better than TAW
0	Same as TAW
-2	Slightly worse than TAW
-5	Significantly worse than TAW

- But how do we pick scores?
 - Anchor scores to evidence (e.g avoid “gut-feel”)
 - Use literature, design textbooks, historical aircraft data, and credible reports
 - “But beware – most people only ever mentions the positives...”
 - Ask
 - “What requirements was this concept solving?”
 - “What part of the concept am I actually scoring?”
 - Novel is not necessarily better
 - Remember – these scores are approximate
 - But try to be consistent across concepts
 - Discuss with teammates to ensure everybody’s 2 or -4 means the same

Number Laundering: MCDA says “C” ...

... but something feels wrong. Why?

- This discomfort is a *signal*, not a failure!
- **MCDA and HoQ are discussion tools**
 - They guard against assumptions, trade-offs and biases
 - But they don’t replace engineering judgement!
- **What Might Have Happened?**
 - Do we have enough engineering criteria?
 - Scalability?
 - How has our HoQ weighted criteria?
 - Or is the process actually telling us something?
- **DO NOT change numbers to game the system**
 - If you don’t think you should add/change criteria, keep the results unchanged
 - Then justify your judgement in the discussion
 - MCDA can be sensitive to individual numbers
 - (e.g., changing 3 to 4 makes B the best)
 - But, I don’t mean changing the number is the correct strategy!
 - Simply the the MCDA chart is there to guide discussion not pick the best concept!

Criteria	Weighting	Concept		
		A	B	C
Mass	0.13	0	-2	0
Drag	0.41	0	3	1
Lift Capability	0.24	0	0	0
Take-off Distance	0.02	0	-1	0
TRL	0.20	0	-3	0
Total		0.00	0.34	0.41
Rank		3	2	1



Previous Example

- A good number of criteria
- But what part of the concept are they grading?
- Weighting was arbitrary*
 - Is cruise speed weighted fairly?

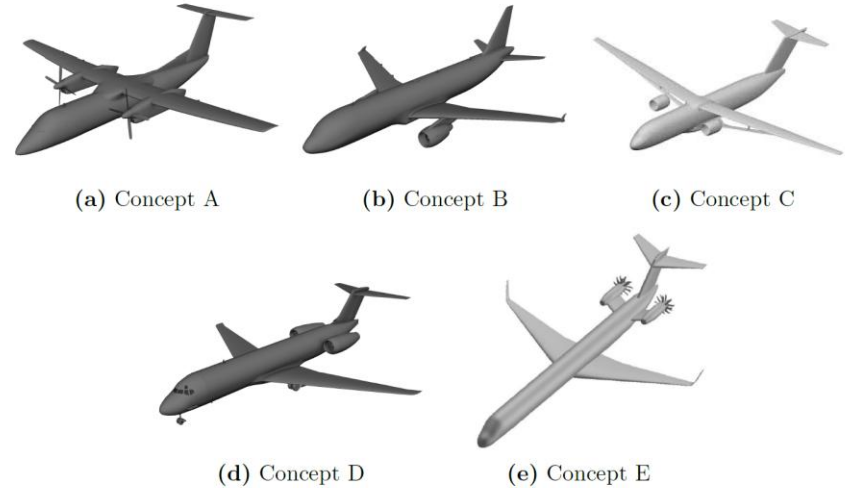


Figure 7: Initial concepts for down selection

Criteria	Weighting	A	B	C	D	E	Scoring	
Development Risk	2	5	5	3	4	3	7	Much better performance than conventional
Manufacturing Costs	2	6	5	3	4	3	6	Better performance than conventional
Fuel Burn	3	7	5	7	5	7	5	Concept B (Conventional)
Turn-Around Time	1	6	5	6	6	6	4	Worse performance than conventional
Cruise Speed	1	4	5	7	6	7	3	Much worse performance than conventional
Technical Maturity	2	5	5	4	5	3		
Stability and Control	1	6	5	5	4	4		
Ease of Maintenance	1	4	5	4	4	4		
Landing Gear Integration	1	4	5	4	5	5		
Total		77	70	67	66	65		
Rank		1	2	3	4	5		

Figure 8: Concept Selection MCDA

Conclusions

Conclusions

- Today, we have explored how to compare multiple configurations under uncertainty
- As a team, you can now compare multiple configurations to choose your “alternative concept.”

Team Submission 1: Concept Down-Selection Review

First (Formative) Team Submission

- Used to help:
 - Guide your teamwork in this first “sprint”.
 - Give you some early feedback
- Most of the content can be reused for your FEDR!
- You’re not “locking yourself in” numbers and values can vary later in the project!
- Please see “*Unit Assessments*” -> “*Briefs*” for the word template!

Team Identity & Executive Summary



Team Identity

Please provide a team name and logo.

Executive Summary

Provide a short executive summary (2–3 sentences) that clearly states the key conclusions from this document. Focus on the main outcomes of your concept down-selection process: the criteria you used, the weighting approach you applied, and the final concepts chosen for further development. This section should give a reader a quick, high-level understanding of your team's decisions without needing to read the full report.

Team Roles

Please complete the following table with the chosen specialism of each team member

Role Name	Team Member
Project Manager / Economics Engineer	
Wing Structures Engineer	
Fuselage Structures Engineer	
Wing Aerodynamics Engineer	
Empennage & Stability Engineer	
Performance, Propulsion & Climate	
Optimisation & Integration Engineer	
Landing Gear Engineer	

Fleet Level Top-Level Aircraft Requirements

This section should include a table of the top-level aircraft requirements (TLARs) you have extracted from the specification. They should be considered at the “Fleet level” (e.g., payload is max. payload transport on each leg).

Provide a 3–5 sentence explanation of how these TLARs may constrain the design space.

Name	Value	Unit
Max. Payload	...	kg
Balanced field length	...	km
Etc.	...	...

Concept Down Selection Methodology

This section will outline the process by which you have chosen your final two concepts, which you will analyse through the MDO development sprint.

Criteria

This section should start with a brief discussion about the critical engineering challenges associated with meeting the specifications requirements.

Then, please list each of the criteria you have chosen to score each concept against. Example criteria are provided for illustration, but they are not necessary to retain. You should aim for between 5 and 15 criteria, depending on what you think is suitable.

The table should also include the relative weighting of each criterion ([with](#) the process used to select these weightings described in the next section).

Note - Ensure criteria are independent and avoid double-counting (e.g., don't include both 'fuel burn' and 'aerodynamic efficiency')

Criteria	Weighting	Brief Description
Readiness of Technology	0.9	How likely is it that all of the required technologies will be ready for the Entry into Service deadline (2040)
Aerodynamic efficiency	0.7	The potential aerodynamic efficiency when compared to a conventional tube-and-wing concept
Etc...		

Criteria weightings

In this section, briefly describe the process used to apply a relative weight to each of your criteria. Any suitable process, such as “AHP”, “House of Quality”, or something of your own choosing/research, is acceptable.

However, remember that the weightings should reflect the customers' requirements rather than your own preferences, and the aim of this section is to explain why each criterion is relevant to the customer requirements or [TLARs.](#)

Include your AHP pairwise matrix or House of Quality mapping as evidence of how weightings were derived, with the final weightings shown in the table above.

Concept Down Selection

Multi-Criteria Decision Analysis (MCDA)

This section should include your final MCDA table. It should include the criteria, their weights, and the score for each concept (at least one concept per team member).

Alternative configuration

This section should start with a brief description of the theme explored across the concepts you chose for comparison. The section should then describe the chosen “alternative configuration” and the key design elements that drove its selection. *(E.g., in your MCDA table, you will have scored your chosen design highly in key criteria. Can you justify these scores?)*

Hyper-parameters

This section should list the chosen hyperparameters you have selected to drive the optimisation process. A description of the selection process is not required. But an acceptable range for each parameter should be provided.

Given the TLARs and an arbitrary value for each of the hyperparameters, your design tools should be deterministic (*e.g for the same value of TLARS and hyper parameters you should get the same “sized” aircraft design*)

Name	Range	Unit
Number of Aircraft	1-1000	-
Fuselage Diameter	2-10	m
Etc...	...	...

Questions?